Crowded Liquidation: Closed-Form Nash and Mean-Field Equilibria of Optimal Execution with Permanent Impact

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Abstract

We extend the Almgren–Chriss optimal liquidation problem to N symmetric traders whose aggregate trading moves the price through linear permanent impact, and to its mean-field limit, in the spirit of Cardaliaguet and Lehalle [2]. Within open-loop strategies we show that every Nash equilibrium is symmetric and solves a single linear ODE in which the impact enters only through the effective coefficient $\alpha(1-1/N)$. This yields closed-form equilibrium trajectories and costs, existence and uniqueness for every $\alpha \in \mathbb{R}$, a proof that crowding ($\alpha > 0$) strictly front-loads liquidation, a closed-form price of anarchy, and $O(1/N)$ convergence to the mean-field limit with an explicit leading constant. All results are verified numerically.

1 Model

Each of N traders $i = 1, \dots, N$ must liquidate an inventory $q_i(0) = Q$ by time T, i.e. $q_i(T) = 0$, trading at rate $\nu_i = q_i'$. Let $\mu(t) = \frac{1}{N}\sum_j \nu_j(t)$ be the average trading rate. The mid-price, cash and objective are

$$dS_t = \alpha\,\mu(t)\,dt + \sigma\,dW_t, \qquad dX_t^i = -\nu_i(S_t + \kappa\nu_i)\,dt, \qquad \max\ \mathbb{E}\big[X_T^i + q_i(T)S_T\big] - \phi\int_0^T q_i^2\,dt,$$

with temporary impact $\kappa > 0$, permanent impact $\alpha \in \mathbb{R}$ and risk penalty $\phi > 0$.

Lemma 1 (Deterministic reduction)**.** *For deterministic (open-loop) strategies, trader i's problem is to minimize*

$$C_i[q_i;\mu] \;=\; \int_0^T \big(\kappa q_i'^2 + \phi q_i^2\big)\,dt \;-\; \alpha\int_0^T q_i\,\mu\,dt.$$

Proof. By Itô's product rule $d(q_iS) = q_i\,dS + S\,dq_i$, so $d(X^i + q_iS) = q_i\,dS - \kappa\nu_i^2\,dt$. Integrating and taking expectations, the dW term has zero mean, giving $\mathbb{E}[X_T^i + q_i(T)S_T] = X_0^i + QS_0 + \alpha\int q_i\mu\,dt - \kappa\int \nu_i^2\,dt$. Subtracting the risk penalty and dropping constants gives $-C_i$. □

Write $k = \sqrt{\phi/\kappa}$ and $I[q] = \int_0^T(\kappa q'^2 + \phi q^2)\,dt$, the Almgren–Chriss functional. Admissible paths are C^1 with $q(0) = Q$, $q(T) = 0$.

2 The single trader

Theorem 1 (Permanent impact is irrelevant to a lone trader)**.** *If $N = 1$ then $C_1[q] = I[q] + \frac{\alpha}{2}Q^2$, and the unique optimal path is*

$$q_{AC}(t) = Q\,\frac{\sinh(k(T-t))}{\sinh(kT)}, \qquad I[q_{AC}] = \kappa k Q^2\coth(kT).$$

Proof. With $\mu = q'$, $\int_0^T qq'\,dt = \big[q^2/2\big]_0^T = -Q^2/2$, a constant. I is strictly convex on the affine space of admissible paths, so its minimizer is unique and is the solution of the Euler–Lagrange equation $\kappa q'' = \phi q$ with the boundary data, which is q_{AC}. For the cost, integrate by parts and use $\kappa q'' = \phi q$: $I[q_{AC}] = [\kappa q q']_0^T = -\kappa Q q'_{AC}(0) = \kappa k Q^2 \coth(kT)$. □

3 Nash and mean-field equilibria

Definition 1. An (open-loop) Nash equilibrium is a profile $(q_1, \dots, q_N)$ such that each q_i minimizes $C_i[\,\cdot\,; \mu]$ given the others' paths. A mean-field equilibrium is a path q that minimizes $C[\,\cdot\,; \mu]$ for a given μ, and satisfies $\mu = q'$.

Lemma 2 (Best response)**.** *Let $m_{-i} = \frac{1}{N}\sum_{j\neq i} \nu_j$. Player i's best response is the unique solution of $2\kappa q_i'' = 2\phi q_i - \alpha\, m_{-i}$ with the boundary data.*

Proof. $C_i = \int \big(\kappa q_i'^2 + \phi q_i^2 - \frac{\alpha}{N} q_i q_i' - \alpha q_i m_{-i}\big)dt$. The term $q_i q_i'$ integrates to a constant, so C_i is strictly convex in q_i and the Euler–Lagrange equation characterizes the unique minimizer. With $L = \kappa q'^2 + \phi q^2 - \frac{\alpha}{N} qq' - \alpha q m$: $\partial_q L = 2\phi q - \frac{\alpha}{N} q' - \alpha m$ and $\frac{d}{dt}\partial_{q'} L = 2\kappa q'' - \frac{\alpha}{N} q'$; equating gives the claim. □

Theorem 2 (Reduction to one ODE)**.** *Every open-loop Nash equilibrium is symmetric, $q_i \equiv q_N$, where*

$$q'' + a\,q' - k^2 q = 0, \qquad q(0) = Q,\ q(T) = 0, \qquad a = a_N := \frac{\alpha}{2\kappa}\Big(1 - \frac{1}{N}\Big). \tag{1}$$

The mean-field equilibrium solves the same problem with $a = a_\infty := \alpha/(2\kappa)$.

Proof. Symmetry. For $i \neq j$ let $d = q_i - q_j$. Since $m_{-i} - m_{-j} = (\nu_j - \nu_i)/N = -d'/N$, Lemma 2 gives $2\kappa d'' - \frac{\alpha}{N} d' - 2\phi d = 0$ with $d(0) = d(T) = 0$. The characteristic roots are real and distinct (discriminant $\alpha^2/N^2 + 16\kappa\phi > 0$), and the only such solution vanishing at both ends is $d \equiv 0$. *Equation.* With all paths equal, $m_{-i} = \frac{N-1}{N} q'$; substituting into Lemma 2 and dividing by 2κ gives (1). *Mean field.* The best response to a given μ is $2\kappa q'' = 2\phi q - \alpha\mu$ (Lemma 2 with m_{-i} replaced by μ, the own-impact term vanishing in the limit). Imposing $\mu = q'$ gives (1) with a_∞. □

Remark 1. A trader's own share $\frac{1}{N}$ of the impact drops out because it only contributes a boundary term. Equilibrium is therefore governed by the impact of *others*, which is why $N = 1$ recovers Almgren–Chriss (Theorem 1).

Theorem 3 (Closed form, existence and uniqueness)**.** *For every $\alpha \in \mathbb{R}$ and $N \in \{1, 2, \dots\} \cup \{\infty\}$, (1) has the unique solution*

$$q_N(t) = Q\, e^{-at/2}\, \frac{\sinh(\omega(T-t))}{\sinh(\omega T)}, \qquad \omega = \omega_N = \sqrt{a^2/4 + k^2} > 0,$$

with cost $I[q_N] = \kappa\,\omega\, Q^2 \coth(\omega T)$.

Proof. Put $q = e^{-at/2} u$. Then $q'' + aq' - k^2 q = e^{-at/2}\big(u'' - \omega^2 u\big)$, so $u'' = \omega^2 u$, whose general solution is $A\sinh(\omega(T-t)) + B\sinh(\omega t)$. The boundary data give $A\sinh(\omega T) = Q$ and $B\sinh(\omega T) = 0$. Since $\omega > 0$, $\sinh(\omega T) \neq 0$, so A, B are uniquely determined ($B = 0$). This holds for every real a, hence every α. For the cost, integrate $\int \kappa q'^2$ by parts and use $\kappa q'' = \phi q - \kappa a q'$: $I[q] = -\kappa Q q'(0) - \frac{\kappa a}{2} Q^2$. With $q'(0) = -Q\big(\frac{a}{2} + \omega\coth(\omega T)\big)$ this equals $\kappa\omega Q^2 \coth(\omega T)$. □

4 Crowding: front-loading and inefficiency

Lemma 3. *$h(x) = x\coth x$ is strictly increasing on $(0,\infty)$.*

Proof. $h'(x) = \dfrac{\sinh x\cosh x - x}{\sinh^2 x} = \dfrac{\frac{1}{2}\sinh(2x) - x}{\sinh^2 x} > 0$, since $\sinh y > y$ for $y > 0$. □

Theorem 4 (Crowding front-loads liquidation)**.** *If $\alpha > 0$ and $N \geq 2$ (including $N = \infty$), then $q_N(t) < q_{AC}(t)$ for all $t \in (0,T)$.*

Proof. Here $a > 0$ and $\omega > k$. With $s = T - t \in (0,T)$,

$$\frac{q_N(t)}{q_{AC}(t)} = e^{-at/2}\cdot\frac{g(\omega)}{g(k)}, \qquad g(w) = \frac{\sinh(ws)}{\sinh(wT)}.$$

The first factor is < 1. For the second, $\frac{d}{dw}\log g = s\coth(ws) - T\coth(wT) = \frac{1}{w}\big(h(ws) - h(wT)\big) < 0$ by Lemma 3, so $g(\omega) < g(k)$. □

Remark 2. For $\alpha < 0$ the factor $e^{-at/2}$ exceeds 1 while $g(\omega)/g(k)$ is still below 1, so the two effects compete. Numerically (Section 6) the exponential dominates for every tested $\alpha \in [-1.5, 0)$ and liquidation is *back*-loaded; a proof is left open.

Theorem 5 (Inefficiency of equilibrium)**.**

1. *A social planner minimizing total cost $\sum_i C_i$ assigns every trader q_{AC}.*
2. *The equilibrium excess cost $\Delta_N = I[q_N] - I[q_{AC}] \geq 0$ for all $\alpha \in \mathbb{R}$, with equality iff $N = 1$ or $\alpha = 0$.*
3. *For $\alpha \geq 0$, the price of anarchy is*

$$\mathrm{PoA}(\alpha, N) = \frac{C[q_N]}{C[q_{AC}]} = \frac{\kappa\omega_N\coth(\omega_N T) + \alpha/2}{\kappa k\coth(kT) + \alpha/2} \;\geq\; 1.$$

Proof. (1) $\sum_i q_i\mu = N\bar{q}\,\bar{q}'$ with $\bar{q} = \frac{1}{N}\sum q_i$, and $\bar{q}(0) = Q$, $\bar{q}(T) = 0$, so $\sum_i C_i = \sum_i I[q_i] + \frac{\alpha N}{2}Q^2$. Each $I[q_i]$ is minimized by q_{AC} (Theorem 1). (2) By Theorems 1 and 3, $I[q_N] = \frac{\kappa Q^2}{T}h(\omega_N T)$ and $I[q_{AC}] = \frac{\kappa Q^2}{T}h(kT)$. Since $\omega_N \geq k$ with equality iff $a_N = 0$, Lemma 3 gives the claim. (3) In a symmetric profile $\mu = q'$, so each trader pays $C = I[q] + \frac{\alpha}{2}Q^2$; divide and use (2). For $\alpha \geq 0$ the denominator is positive so the ratio is meaningful. □

Remark 3. For sufficiently negative α (here $\alpha < -2\kappa k\coth(kT) \approx -0.63$) total costs become negative and the ratio loses meaning; the excess cost Δ_N is the right measure.

5 Convergence to the mean-field limit

Theorem 6 ($O(1/N)$ convergence with explicit constant)**.** *Write $q(t;a)$ for the solution in Theorem 3. Then*

$$\big\|q_N - q_\infty\big\|_\infty = \frac{|\alpha|}{2\kappa N}\,\big\|\partial_a q(\cdot;a_\infty)\big\|_\infty + O(N^{-2}),$$

where $\partial_a q(t;a) = q(t;a)\Big[-\frac{t}{2} + \frac{a}{4\omega}\big((T-t)\coth(\omega(T-t)) - T\coth(\omega T)\big)\Big]$.

Proof. $a_N - a_\infty = -\alpha/(2\kappa N)$. The map $(t,a) \mapsto q(t;a)$ is smooth on $[0,T]\times\mathbb{R}$ (the bracket extends continuously to $t = T$), so Taylor's theorem in a with $\partial_a^2 q$ bounded on the compact set $[0,T]\times[a_N, a_\infty]$ gives $q_N - q_\infty = (a_N - a_\infty)\partial_a q(\cdot;a_\infty) + O(N^{-2})$ uniformly in t. The formula follows from $\partial_a \log q = -\frac{t}{2} + \frac{d\omega}{da}\big((T-t)\coth(\omega(T-t)) - T\coth(\omega T)\big)$ and $d\omega/da = a/(4\omega)$. □

6 Numerical verification

Parameters: $T = Q = 1$, $\kappa = 0.1$, $\phi = 1$ (so $k = \sqrt{10}$). Code: `src/mfg.py`.

Check	Result	Theorem
Closed form vs. `solve_bvp`, $\alpha \in [-1.5, 5]$, $N \in \{1, 2, 10, 100, \infty\}$	max rel. error 5.8×10^{-14}	3
$q_1 = q_{AC}$	difference exactly 0	1
Inventory sold by $T/2$: AC / α=1 / α=5 / α=-1	80.3% / 96.2% / $\approx 100\%$ / 54.3%	4
$\mathrm{PoA}(\alpha, \infty)$ at $\alpha = 1$, 2, 5	1.105, 1.208, 1.345	5
Fitted slope of $\log \lVert q_N - q_\infty \rVert_\infty$ vs. $\log N$, $N \in [10, 10^4]$	-1.002	6
$N \lVert q_N - q_\infty \rVert_\infty$ at $N = 10^6$ vs. predicted constant ($\alpha = 1$)	0.2272876 vs. 0.2272875	6

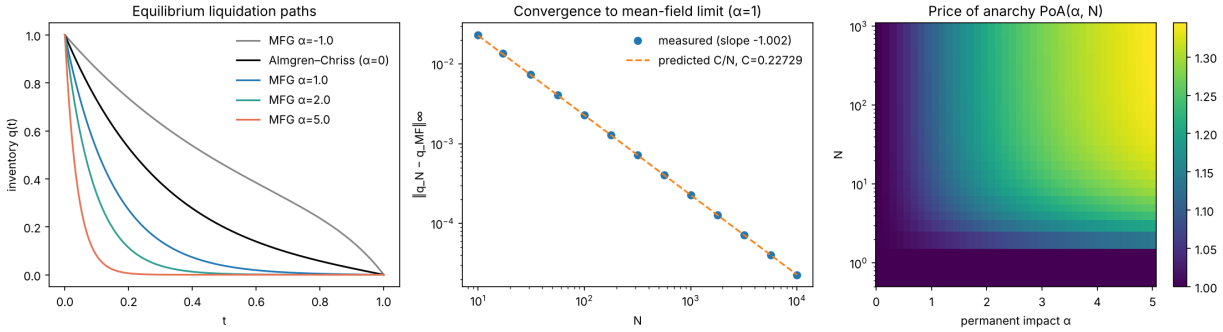


Figure 1: Left: equilibrium paths; crowding ($\alpha > 0$) front-loads, $\alpha < 0$ back-loads. Centre: $O(1/N)$ convergence, measured vs. predicted constant. Right: price of anarchy.

7 Limitations and extensions

Strategies are open-loop and deterministic; closed-loop (feedback) Nash equilibria of the N-player game generally differ at finite N and are characterized by coupled Riccati equations. Impact is linear and traders are homogeneous. Natural extensions are heterogeneous initial inventories, which lead to a forward–backward ODE for the mean inventory as in [2], and a proof of back-loading for $\alpha < 0$.

References

[1] R. Almgren and N. Chriss. Optimal execution of portfolio transactions. *Journal of Risk*, 3(2):5–39, 2001.

[2] P. Cardaliaguet and C.-A. Lehalle. Mean field game of controls and an application to trade crowding. *Mathematics and Financial Economics*, 12:335–363, 2018.